

Thermodynamics

Thermal equilibrium :- The two systems are said to be in thermal equilibrium with each other if they are, at the same temp.

Thermodynamic system :- The assembly of extremely large no. of atoms & molecules or particles which have a certain value of pressure, volume & temp. is called a thermodynamic system.

Thermodynamic variable :- The variables which determine the thermodynamic behaviour of a thermodynamic system are called thermodynamic variables.

Ex - Pressure (P), Volume (V), Temp (T)
internal energy (U)

Equation of state :- The equation which shows the relation between P , V , & T is called equation of state.

(i) $P \propto \frac{1}{V} \rightarrow \text{Boyle's law} \Rightarrow V \propto \frac{1}{P}$

(ii) $V \propto T \rightarrow \text{Charles law}$

$$V \propto T/P$$

$$PV \propto T$$

$$PV = RT \text{ or } PV = NRT \quad (N=1, \text{ mole}=1)$$

where R = gas constant

$$\star \quad R = \frac{PV}{T} = \frac{Nm^{-2}m^3}{K} = Nm K^{-1} = JK^{-1}mol^{-1}$$

$$\star \quad R = 8.31 J mol^{-1} K^{-1}$$

Thermodynamic process - A process is said to be taking place if the thermodynamic variable of the system changes with time.

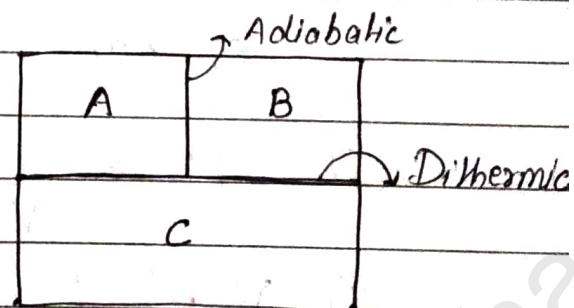
Types of thermodynamic process :-

- (i) Isothermal process :- Temperature $\rightarrow$ const
- (ii) Isochoric process :- Volume $\rightarrow$ const
- (iii) Isobaric process :- Pressure $\rightarrow$ const
- (iv) Adiabatic process :- Heat (Q) $\rightarrow$ const
- (v) Cyclic process :- System returns to its original state.

Zeroth law of thermodynamics

If two systems A and B are in thermal equilibrium with the system C then the system A and B will be in thermal equilibrium.

Adiabatic (insulating)
do not allow to flow heat



Diathermic (conducting)
allow to flow heat

Internal energy

In case of ideal gases (perfect gases) the molecules do not attract each other but it is not true in actual practice where there is so, a gas is called real gas.

Therefore, the molecules of a real gas possess intermolecular potential energy. In such a situation the volume of gas increases, work will be performed by the gas against intermolecular attraction & hence the potential energy of the gas increases.

Expansion - internal
P.E increases

Contraction - internal P.E
decreases

As a whole internal energy of the thermodynamic system containing gas is the result of

- 1) Internal P.E which is the function of volume
- 2) K.E which is the function of temp.

$$U = I.P.E + K.E$$

For ideal gas $I.P.E = 0$

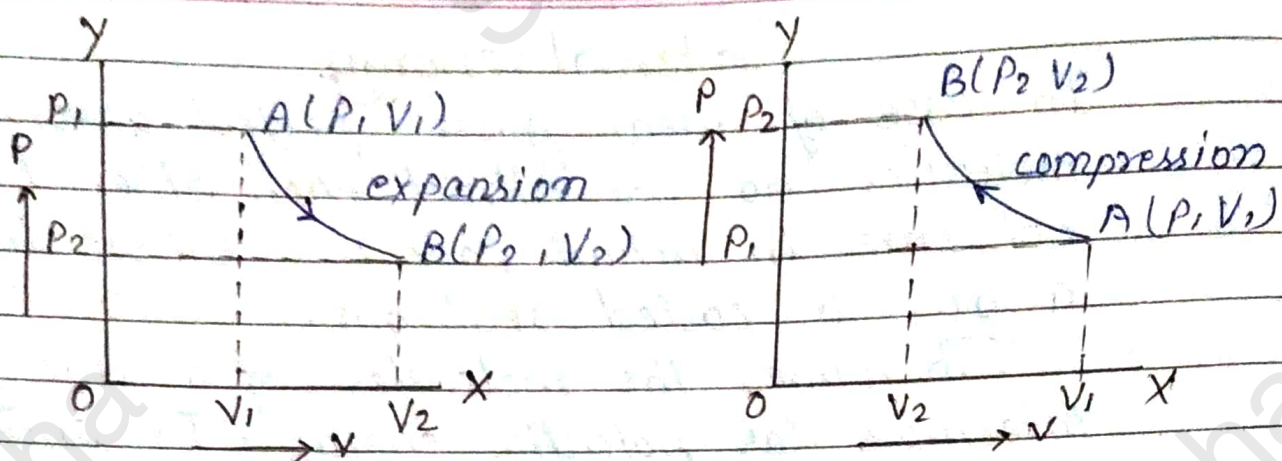
$$U = K.E$$

Indicator diagram (P-V diagram)

The graphical representation of the state of a system with the help of two thermodynamical variables is called indicator diagram of the system.

It is of two types :-

- 1) For compression
- 2) For expansion



Work done during expansion / compression of gas.

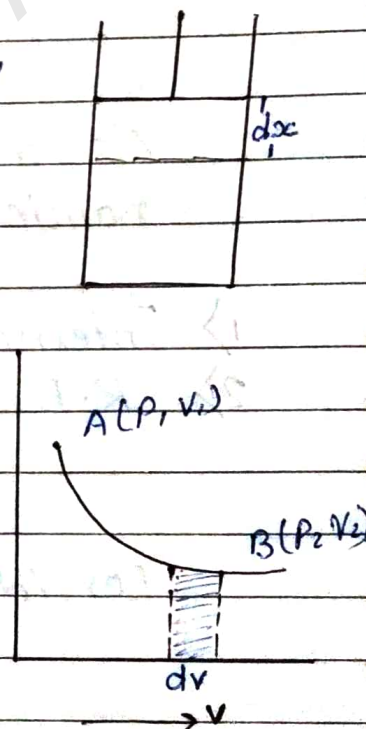
Suppose that a cylinder with movable piston contains a gas of 1 mole at pressure P & volume V .

If the piston of CSA is compressed with a small displacement dx then force exerted on the gas.

$$F = Pa$$

Now, small work done by the piston on the system.

$$\begin{aligned} dw &= F \cdot dx \\ dw &= P a dx \quad \left\{ \begin{array}{l} a dx = \text{small} \\ \text{change in volume} \end{array} \right. \\ dw &= P dv \end{aligned}$$



$dw = \text{area under the } P-V \text{ diagram}$

$$\text{Total work done} = \int dw = \int_{v_1}^{v_2} P dv$$

Pressure can be assumed constant

$$W = \int_{v_1}^{v_2} P dv$$

In case of isothermal process

$$W_{iso} = \int_{v_1}^{v_2} P dv$$

$$W_{iso} = \int_{v_1}^{v_2} \frac{RT}{v} dv$$

$$W_{iso} = RT \int_{v_1}^{v_2} \frac{1}{v} dv$$

$$W_{iso} = RT [\log v]_{v_1}^{v_2}$$

$$W_{iso} = RT [\log v_2 - \log v_1]$$

$$W_{iso} = RT [\log e \cdot \frac{v_2}{v_1}]$$

$$W_{iso} = RT \left[\log e \frac{v_2}{v_1} \right] = 2.303 RT \log e \frac{v_2}{v_1}$$

$$W_{iso} = RT \left[\log e \frac{P_1}{P_2} \right] \quad \left\{ \begin{array}{l} \because P_1 v_1 = P_2 v_2 \\ \text{So,} \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{v_2}{v_1} = \frac{P_1}{P_2} \end{array} \right.$$

In case of adiabatic process

$$W_{adia} = \int_{v_1}^{v_2} P dv$$

$$W_{adia} = \int_{v_1}^{v_2} \frac{K}{v^Y} dv$$

$$W_{adia} = K \int_{v_1}^{v_2} v^{-Y} dv$$

$$W_{adia} = K \left[\frac{v^{-Y+1}}{-Y+1} \right]_{v_1}^{v_2}$$

$$W_{adia} = \frac{K}{1-Y} \left[v_2^{1-Y} - v_1^{1-Y} \right]$$

$$W_{adia} = \frac{K}{1-Y} \left[v_2^{1-Y} - v_1^{1-Y} \right]$$

$$W_{adia} = \frac{1}{Y-1} \left[K v_1^{1-Y} - K v_2^{1-Y} \right]$$

$$W_{adia} = \frac{1}{Y-1} \left[P_1 v_1^Y v_1^{1-Y} - P_2 v_2^Y v_2^{1-Y} \right]$$

In adiabatic process

$$P v^Y = K$$

where Y is equal to $\frac{C_p}{C_v}$

$$P = \frac{K}{v^Y}$$

$$P v^Y = K$$

$$\text{So, } P_1 v_1^Y = P_2 v_2^Y = K$$

$$W_{\text{adia}} = \frac{1}{\gamma - 1} [P_1 V_1 - P_2 V_2]$$

or

$$W_{\text{adia}} = \frac{R}{\gamma - 1} [T_1 - T_2]$$

First law of thermodynamics

If an amount of heat is given to a thermodynamic system, then a part of heat causes increase in its internal energy and the remaining part causes amount of work done by the system.

$$\text{i.e., } dQ = du + dW$$

This law is just another form of the law of conservation of energy.

$$dQ = du + pdv \quad \{dW = pdv\}$$

(i) In case of isothermal process

$$T \rightarrow \text{const or } du = 0$$

$$\therefore dQ = dW$$

It means in this process the amount of heat given to the system will be consumed or used up to perform work by the system.

- $dQ (+ve) \rightarrow$ heat given $\rightarrow dW (+ve)$ work done by the system
- $dQ (-ve) \rightarrow$ heat extracted $\rightarrow dW (-ve)$ work done by the system

(ii) In case of adiabatic process

$$Q \rightarrow \text{const, } dQ = 0$$

$$dQ = du + dW$$

$$dW = -du$$

- dw (+ve) work done by the system $\rightarrow du$ (-ve) temp. decrease
- dw (-ve) work done on the system $\rightarrow du$ (+ve) temp. increases

(iii) In case of isochronic process

$$v = \text{const} \quad \text{i.e. } dv = 0$$

$$dQ = du + dw$$

$$dQ = du + PdV \quad \{ PdV = 0 \}$$

$$dQ = du$$

Specific heat of gases

In Specific heat - It is the amount of heat which is required to increase the temp of a substance of 1 kg through 1°C .

In case of solid & liquids there is negligible change in the pressure & volume of the substance raising the temp. through 1°C . So, such a substance has only 1 value of specific heat.

But in case of gases when temp is raised through 1°C , pressure & volume both changes. So, the specific heat of gas is studied once keeping the pressure const & another keeping the volume constant.

Specific heat of gas at const pressure (C_p)

Amount of heat required to increase the temp. of a gas of unit mass through 1°C at const pressure.

Specific heat of gas at const volume (C_v)

Amount of heat required to increase the temp. of a gas of unit mass through 1°C at const volume.

$$SI \text{ unit} = J K^{-1} kg^{-1}$$

$$\therefore Q = ms\theta \Rightarrow s = \frac{Q}{m\theta}$$

$$C_v = \frac{Q}{m\Delta T} \quad C_p = \frac{Q}{m\Delta T}$$

Molar Specific heat capacities of Gases

Amount of heat required to raise the temp of 1 mole of a gas by $1^\circ C$

$$\text{Molar specific heat at const volume } C_v = \frac{Q}{n\Delta T}$$

$$\text{Molar specific heat of const pressure } C_p = \frac{Q}{n\Delta T}$$

$$C_v = M c_v$$

$$C_p = M c_p \text{ where } M \text{ is molar mass}$$

Q Why C_p is greater than C_v ?

When a gas is heated to increase the temp. through $1^\circ C$, both volume & pressure change. This is why a gas consists of two values of specific heats C_p & C_v out of which C_p is greater than C_v which can be explained as below.

(i) At constant volume

In such a situation the amount of heat supplied to the gas is completely used up to increase the temperature of the gas through $1^\circ C$ as no work is done.

(ii) At const pressure

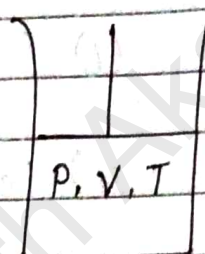
In such a situation the same amount of heat as earlier is partially used up to perform the work along with the increase in its temp.

Clearly, this amount of heat will not be able to increase the temp of gas through 1°C .

Thus to increase the temp through 1°C at const pressure the gas needs more amount of heat i.e. $C_p > C_v$

Relation between C_p & C_v

There is a cylinder with movable piston containing 1 mole of a gas at pressure P , volume V & temp T



If C_v is the specific heat of gas at const V then heat required to increase the temp of the gas through dT ,

$$dQ = 1 C_v dT \text{ or } C_v dT \quad \{ Q = m s \theta \} \quad \text{--- (1)}$$

If C_p is the sp heat of gas at const P , then the heat required to increase the temp of the gas through dT

$$dQ' = C_p dT \quad \text{--- (2)}$$

$$\therefore dQ' > dQ$$

$$dQ' = dQ + dW$$

$$C_p dT = C_v dT + dW$$

$$dT (C_p - C_v) = p dv \quad \text{--- (3)}$$

From eq of state, $PV = RT$

differentiating with respect to T

$$P \cdot \frac{dv}{dT} = R \times 1 \times T^{-1}$$

$$\Rightarrow P dv = R T dT \quad \text{--- (4)}$$

Putting (4) in (3)

$$dT (C_p - C_v) = R dT$$

$$C_p - C_v = R$$

or $C_p - C_v = \frac{R}{J}$ (for $R = \text{work}$) where
 $J = \text{conversion factor}$, $J = \text{mechanical equivalent of heat}$.

Also, $\frac{C_p}{C_v} = \gamma$, γ represent the atomicity of the gas.

(i) For monoatomic gas (He, Ar, Kr, Xe, Rn)

$$\gamma = 1.33 \text{ } 1.67$$

(ii) For diatomic gases (H_2 , O_2 , N_2)

$$\gamma = 1.41$$

(iii) For triatomic gases (O_3 , SO_2 , CO_2)

$$\gamma = 1.67 \text{ } 1.33$$

Differences between isothermal & adiabatic process

Isothermal	Adiabatic
1> Temperature constant	Heat constant
2> $dT = 0$	$dQ = 0$
3> Takes place in cylinder of conducting walls	Takes place in cylinder of insulating walls.
4> Slow process	Fast process
5> Ex - melting of ice, Boiling of water, leakage of gas in tube of tyre	Ex - Bursting of tube of a tyre.

Equation of state of adiabatic process

$$P V^\gamma = \text{const}, \text{ where } \gamma = C_p / C_v$$

$$TV^{\gamma-1} = \text{Const}$$

$$p^{1-\gamma} T^{\gamma} = \text{Const}$$

2nd law of thermodynamics

This law comes into existence when the following 3 limitations come into light :-

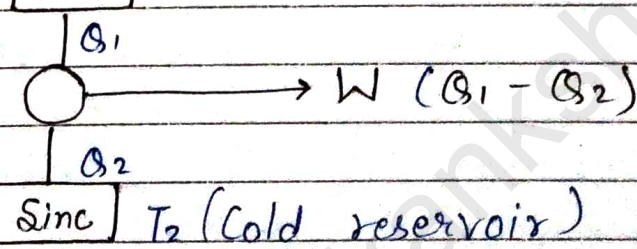
- (i) 1st law does not indicate the direction of heat transfer.
- (ii) It does not indicate the conditions under which heat can be transferred into work.
- (iii) It does not indicate why heat cannot be converted completely into mechanical work.

Kelvin's statement of 2nd law of thermodynamics :-
It is impossible for any device as heat engine that to operates on a cycle to receive heat from a single reservoir and produce net amount of work.

Clausius statement of 2nd law of thermodynamics :-
It is impossible to construct a device as heat pump that operates in a cycle and produces no effect other than the transfer of heat from lower temperature to higher temperature body.

Result :- "Heat cannot flow itself from colder body to a hotter body,

Here, the work substance T_1 [Source] $T_1 > T_2$
absorbs an amount of heat Q_1
from a source at temp T_1 &
converts its a part into
mechanical work (W) [Sink] T_2 (Cold reservoir)



rejecting amount of heat Q_2 to the sink at temp T_2 ($T_1 > T_2$)

Clearly $W = Q_1 - Q_2$

Efficiency of heat engine

$$\boxed{\eta = \frac{W}{Q_1}} = 1 - \frac{Q_2}{Q_1}$$

If $Q_2 = 0$, $\eta = 100\%$

$$\eta \% = 1 - \frac{Q_2}{Q_1} \times 100$$